

CLASH OF THE **CLAWS**

RVCE _ ClawGrowth

*Drafts everything. Sends nothing.
Until you say yes.*

OpenClaw Venture Partner - An autonomous AI deal-flow agent for freelancers and small agencies.

Sathish - Team Lead



Problem Statement

Theme: GenAI - Productivity

- Freelancing is repetitive scaffolding work. LLMs were built to absorb exactly this layer - but every shipped tool fully automates and breaks the trust loop.

The Pain (4 - 8 hrs / week wasted)

- **Manual scraping** - checking 5+ job boards daily, deduping leads.
- **Cold outreach** - copy-paste templates that get 1 - 2 % reply rates.
- **Reply triage** - genuine interest vs. polite no vs. scope question.
- **Context-switching** - Gmail, Notion, Trello, Slack - no memory between them.

Persona

- Solo freelancers and 2 - 5 person agencies hunting on Upwork, LinkedIn, Indeed, Reddit, Contra and Freelancer.

Why it matters

- Full autopilot is dangerous: one wrong email to a real client kills the relationship. Operators want a draft-everywhere, approve-on-glance partner - not autopilot.

4 - 8

hours / week wasted on lead
scrubbing

1 - 2 %

typical cold-outreach reply rate

*Operators don't want autopilot.
They want a co-pilot they can trust.*

Current Solution & Gaps

What's already out there

- **Generic AI writers** - Jasper, Copy.ai , Spit out templates, no domain awareness.
- **Outreach autopilots** - Lemlist, Apollo, Instantly. Auto-send first; ask later.
- **Job-board scrapers** - single-source, brittle, no scoring or memory.
- **SaaS CRMs** - Notion, HubSpot. Storage, not intelligence.

What's missing / broken

- No tool ties scout, pitch, negotiate and memory into one loop.
- Auto-send is the default - one bad email burns a real client.
- Costs USD 30 - 200 / month before any value is proven.
- Closed-source. You can't extend, audit or self-host.

The opportunity

- An open-source, self-hosted, draft-only operator partner that scouts, drafts and negotiates - but never sends until a human says yes.

Comparison snapshot

- **Today's tools** - auto-send first, single-source scrape, paid SaaS, closed.
- **OpenClaw VP** - draft-only, 6 sources, \$0 budget, open-source, self-hosted, persistent client memory.

Why now

- Free-tier LLMs are now strong enough for nuanced classification and drafting. Always-Free cloud tiers (Oracle Cloud, Vercel, Supabase) make a 24 / 7 agent runtime possible at zero cost - the entire stack runs without a card.

Our Solution - 3-Layer Agent

What we built

- An autonomous deal-flow agent for freelancers and small agencies that runs in the background and continuously does three things - **scout, draft, negotiate** - while every outbound stays gated behind a human approval.

Core idea

- **Layer 1 - SCOUT.** Pulls leads from 6 boards every hour, scores 0 - 100, auto-pitches scores 80+.
- **Layer 2 - ARCHITECT.** Attaches a real proof artifact (Lighthouse audit, code sample, design teardown) to high-confidence pitches.
- **Layer 3 - NEGOTIATOR.** Classifies every inbound reply, drafts 3 tone-variant responses, accumulates a per-client memory.md.

Key features

- Two surfaces, one state - web dashboard and Telegram / Discord chat both share live state via Supabase Realtime.
- Brand promise - **draft-only**. The agent never sends anything autonomously.

The loop

SCOUT leads in -> scored 0 - 100

Upwork - LinkedIn - Indeed - Reddit - Contra - Freelancer

▼

ARCHITECT pitch + proof attached

Lighthouse audit, code sample, design teardown

▼

NEGOTIATOR classify reply -> 3 tone drafts

positive / negative / question / unsubscribe

▼

HUMAN APPROVAL = GREEN BUTTON

Every outbound = HITL gate

Inside the 3 Layers

Layer 1 - SCOUT

- **What it does:** scrapes 6 boards every hour - Upwork, LinkedIn, Indeed, Reddit, Contra, Freelancer.
- **Scoring:** LLM scores each lead 0 - 100 on stack match, budget fit, signal strength.
- **Auto-pitch:** top 5 leads with score 80+ kick off pitch drafts via Inngest events.
- **Resilience:** per-source token-bucket rate limits, exponential-backoff retry (1s / 4s / 16s), Zyte -> Firecrawl -> stub adapter cascade.

Layer 2 - ARCHITECT

- **What it does:** builds a real proof-of-value artifact for high-confidence pitches.
- **Examples:** Lighthouse audit on the prospect's site (PageSpeed Insights API), code sample, design teardown.
- **Why it works:** the artifact becomes the pitch's hook - proof beats prose every time.

Lead Inbox
10 leads queued · sorted by score

Search leads, clients, pitches

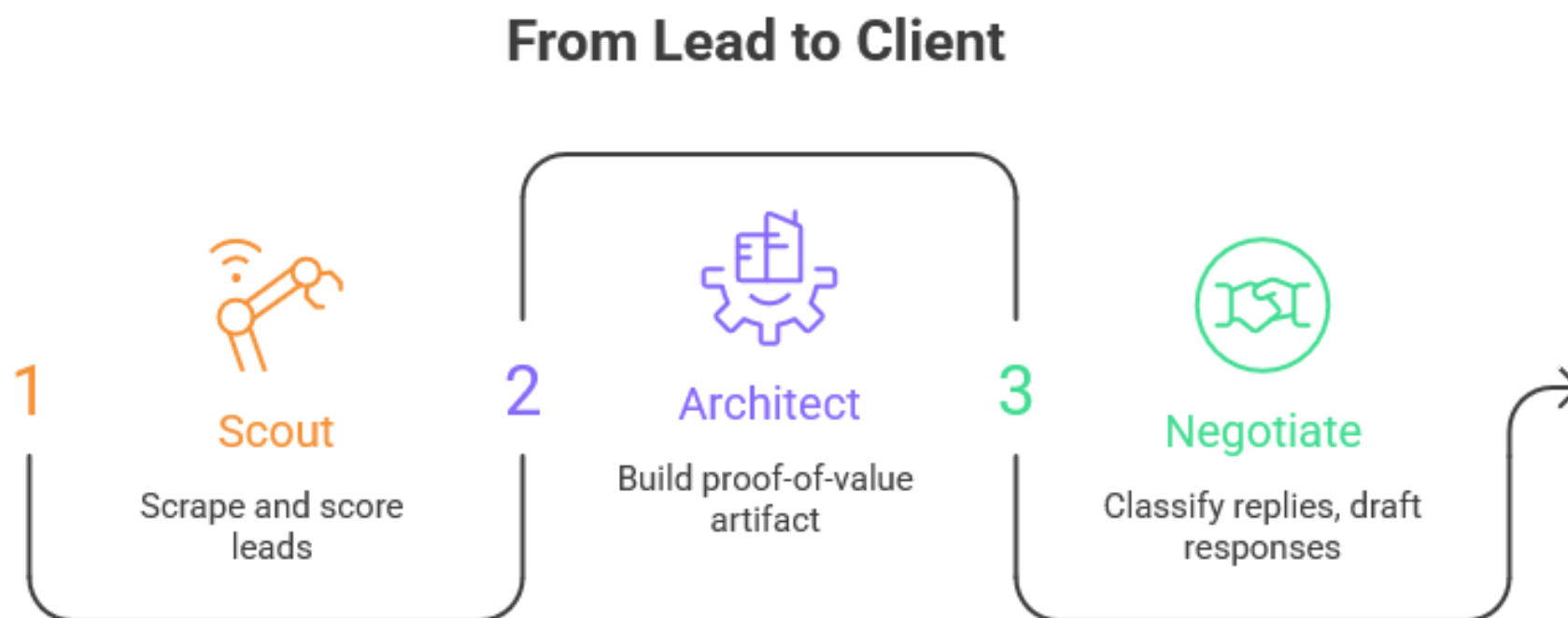
react next.js

Run scout

LEADS QUEUED	PITCHES SENT	REPLY RATE	HOURS SAVED
10	0	—	—

SCORE	LEAD	SOURCE	LAYER	AGE	STATUS
94	Senior Next.js engineer for SaaS an...	c9912a7c · \$6,200 · Upwork	L1	now ago	DRAFT READY
89	Next.js + Sanity portfolio site for des...	0b12e3e9 · \$2,800 · Contra	L1	now ago	DRAFT READY
85	Tailwind component library polish + ...	cb8d3a52 · \$3,800 · Upwork	L1	now ago	DRAFT READY
82	React/Redux dashboard rewri...	8a55bb8f · \$4,200 fixed · Upwork	L1	now ago	DRAFT READY
81	Shopify Hydrogen migration contrac...	07a38db9 · \$75/hr · Indeed	L1	now ago	DRAFT READY
66	[Hiring] React Native dev for M...	3cfa7a94 · \$2,000 flat · Reddit	L1	now ago	DRAFT READY
45	Looking for engineers to help build somet...	a9f422fd · — · LinkedIn	L1	now ago	DRAFT READY
30	Frontend Lead -- Series B fintec...	bba17d57 · \$135k-160k · LinkedIn	L1	now ago	DRAFT READY
28	Junior React developer wanted, 10 h...	b9e98a6b · \$15/hr · Upwork	L1	now ago	DRAFT READY
22	PHP / Laravel developer for legacy ...	55cd557e · \$40/hr · Indeed	L1	now ago	DRAFT READY

Inside the 3 Layers



Layer 3 - NEGOTIATOR

- **Inbound classifier:** every reply tagged as positive / negative / question / unsubscribe.
- **3 tone-variant drafts:** operator picks the best fit - formal, casual, decisive - and approves.
- **Client memory.md:** once a deal is won, every interaction (preferences, signals, last context) appends to a per-client markdown blob the next pitch quietly reads.
- **Daily upsell cron:** scans active clients for upsell signals once a day, drops drafted nudges into the queue.

The HITL gate

- Every layer ends in a draft, never a send. The operator approves from the web dashboard, Telegram inline keyboard, or Discord embed button - all three taps hit the same handler and broadcast the same Realtime event.

Drafts everywhere. Sends only on green.

Product Walkthrough - 90 seconds

The flow a judge sees

1. Open **/agent** - green Gateway-connected card, OpenClaw Control UI loads in an iframe.
2. Type "**find leads for next.js**" - Gateway matches the scout skill, calls the runScout MCP tool, fires an Inngest event.
3. Switch to **/inbox** - leads stream in live via Realtime as they're scraped and scored.
4. Top lead at **score 92** - the auto-pitch worker has already drafted a body via the streaming helper.
5. Open the **pitch** - see the body, the proof artifact (Lighthouse 85 -> 95), the LLM reasoning.
6. Approve from **web** - instant flip to "sent"; check Resend dashboard for the actual email.
7. Or approve from **Telegram** - inline keyboard, one tap, dashboard updates simultaneously.
8. **Inbound reply** (simulated via /api/email/simulate) - classifier runs, three drafts appear.
9. Pick the best draft, approve - Resend sends, clients.memory_md gets a fresh entry appended.

The wow moment

Approve from any surface -
Web button | Telegram tap |
Discord embed

All three update each other **live**
via Supabase Realtime.

30 sec

per pitch from glance to send

*The agent works around the clock.
The operator approves on glance.*

Tech Stack & System Flow

Frontend & runtime

- **Next.js 14 App Router on Vercel** - SSR + edge, free tier covers prototype.
- **Inngest serverless workers** - event-driven, retries, dedupe via idempotency keys.
- **TypeScript strict, monorepo** - apps/web + apps/worker + apps/agent + 5 packages, ~25K LOC.

Data & state

- **Supabase Postgres** - 15 tables, RLS, Auth, Realtime in one platform.
- **Upstash Redis** - rate limiting, idempotency, caching.

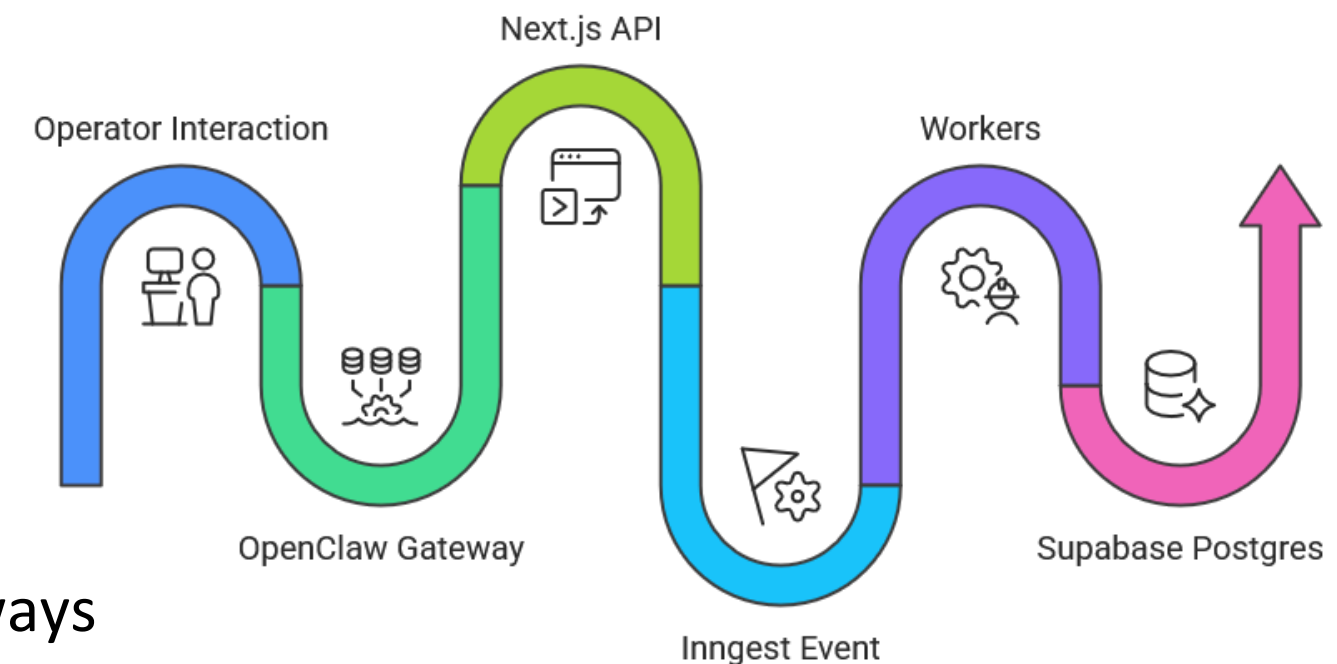
AI & agent

- **OpenClaw Gateway** - open-source skill-based agent host on Oracle Cloud Always Free.
- **5-LLM cascade** - Copilot -> Gemini -> Groq -> OpenRouter -> Anthropic.
- **Scraping** - Zyte (primary) + Firecrawl (secondary), JSON-LD JobPosting fallback for LinkedIn / Freelancer.

Email & channels

- Resend for outbound. Telegram + Discord + Slack via Gateway. WhatsApp planned.

System Flow



OpenClaw Integration

OpenClaw is not a library - it's a deployable Gateway process. We integrate via three surfaces:

9 production skills

- Markdown files with YAML frontmatter the Gateway loads on startup. Each maps a natural-language request to MCP tool calls.
- **scout → draft_pitch → approve_pitch → reject_pitch → show_top_lead → client_memory → reply_to_email → lighthouse_audit → help**

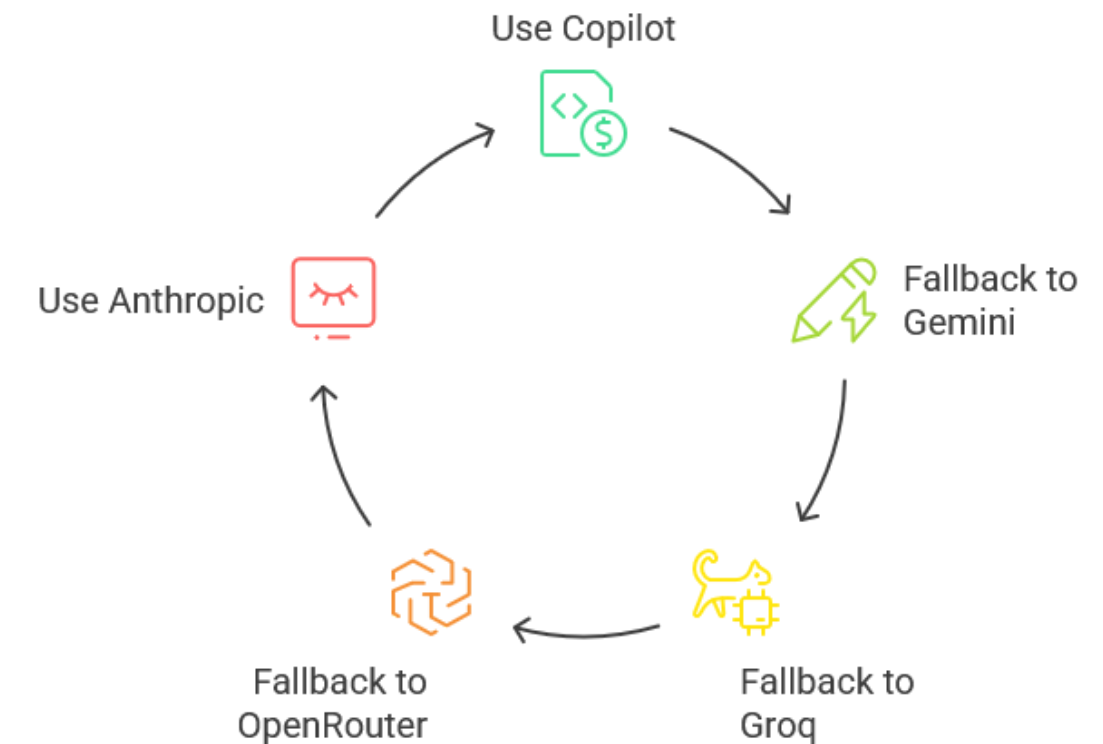
11 MCP tools

- JSON-RPC over HTTP. Authenticated with **MCP_SHARED_SECRET** (Gateway) and **MCP_WORKER_SECRET** (workers) → **separate trust tiers**.
- **runScout → getRecentLeads → getTopLead → draftPitch → getPendingPitches → approvePitch → rejectPitch → editPitch → bindTelegram → bindDiscord → notifyAgent**

Control UI iframe

- The Gateway hosts its own chat UI; we embed it as an iframe on /agent so operators get conversational chat *inside* our dashboard. No context-switch, same skills, same tools.

AI Model Selection Cycle



Budget guard: USER_DAILY_BUDGET_USD env, BudgetExceededError throw, idempotency keys, response caching.

Channels: Telegram + Discord live; Slack ready; WhatsApp queued (Meta verification pending).

HITL, Data & Security

15 tables, all RLS-locked

- profiles → sources → leads → scores → pitches → clients → approvals → llm_calls → scrape_failures → provider_health → audit_log → email_replies → proof_artifacts → scout_runs → binding_codes
- Materialized view **user_daily_spend** drives the budget guard. 17 SQL migrations, applied in order.

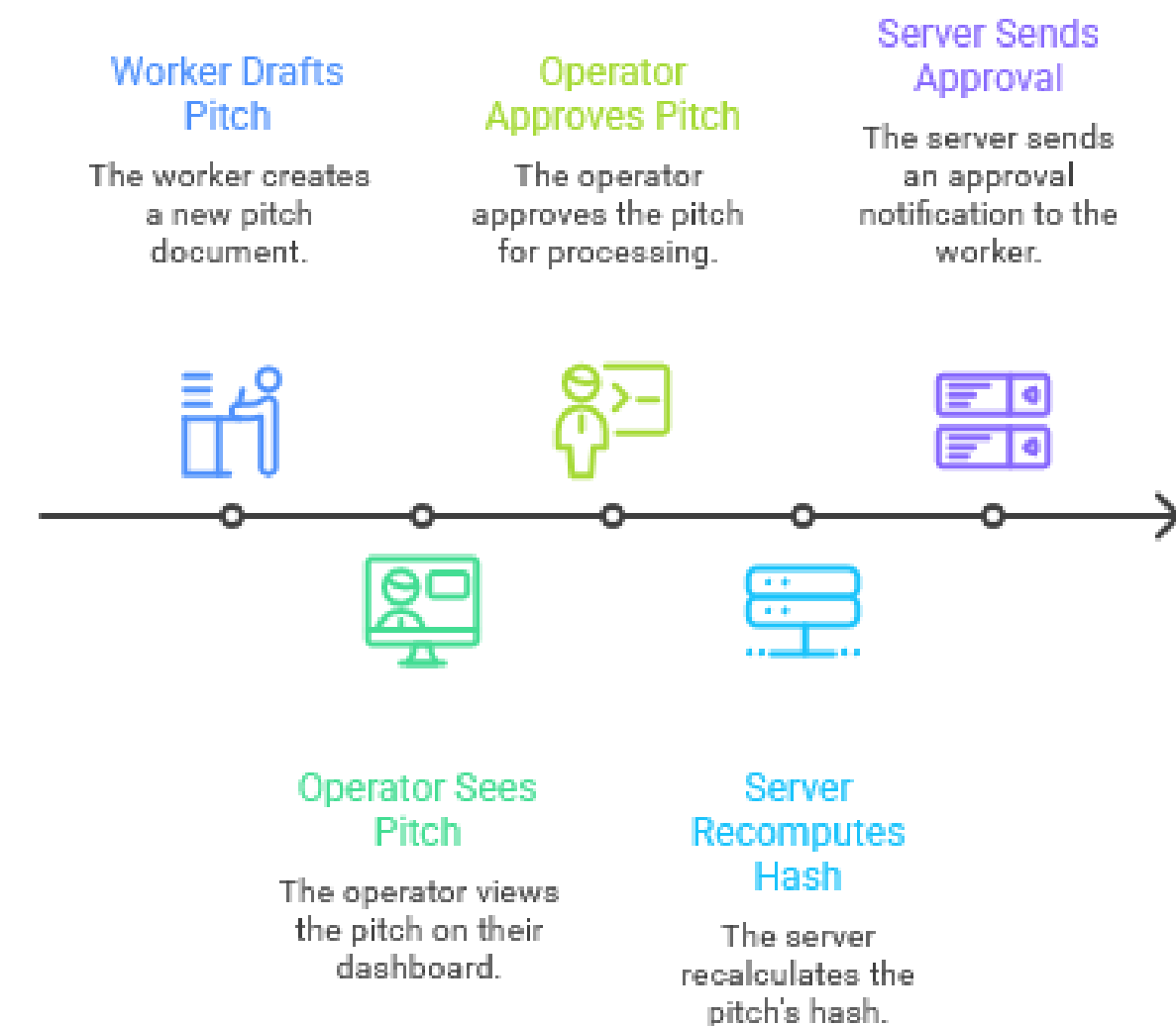
HITL - the cornerstone

- Every outbound email rides through a hash-bound approval gate.
- **payload_hash = SHA-256(subject + body + lead_id)**
- Approval handler recomputes the hash from current content. If it doesn't match what the operator saw, server returns **409 Conflict** and refuses to send.

Security stack

- **RLS** - users only see their own rows.
- **Auth** - Supabase email + magic link, validated by Next.js middleware.
- **Rate limits** - Upstash Redis (in-memory fallback). /api/scout 10/hr; pitches 30/day.
- **Audit log** - append-only, every sensitive action: actor / action / target / timestamp.
- **Two MCP secrets** - Gateway and worker callbacks held to separate trust tiers.

HITL Approval Flow



If anyone edits the pitch between approval and send, the hash check breaks. Approval is bound to the exact content the operator saw.

Impact & Use Cases

Real-life scenarios

1. **Solo Next.js dev, Bangalore** - scout queries "next.js + tailwind" overnight; wakes to 4 scored leads, 2 with Lighthouse audits attached. Approves 2 in under 60 seconds.
2. **3-person design agency** - each member binds their Telegram; agent broadcasts new pitches to a shared queue; whoever's awake taps Approve.
3. **WordPress dev hunting Reddit** - scout pulls r/forhire posts with "WordPress" + "budget"; classifier filters out polite no-thanks; only positive replies surface.
4. **Returning client upsell** - daily upsell-detector cron scans clients.memory_md, finds a stale customer who mentioned a redesign 3 months ago, drafts a check-in nudge.

Quantified impact (per operator)

- 4 - 8 hours / week reclaimed from manual lead scrubbing.
- Lead volume scales with the number of source-board parsers (today = 6: Upwork, LinkedIn, Indeed, Reddit, Contra, Freelancer).
- Pitch quality lifted by the Layer-2 proof artifact (Lighthouse audit, code sample, design teardown) attached at score 80+.
- Zero out-of-pocket - budget guard ensures no LLM spend ever exceeds the USER_DAILY_BUDGET_USD env ceiling.
- Hours-saved is tracked in the dashboard from approved-pitch counts and a per-action heuristic (Phase 6).

Scale potential

\$0

onboarding cost - no card required

~45 min

operator setup, end-to-end

From repo clone to first approved pitch.

Differentiation & MOAT

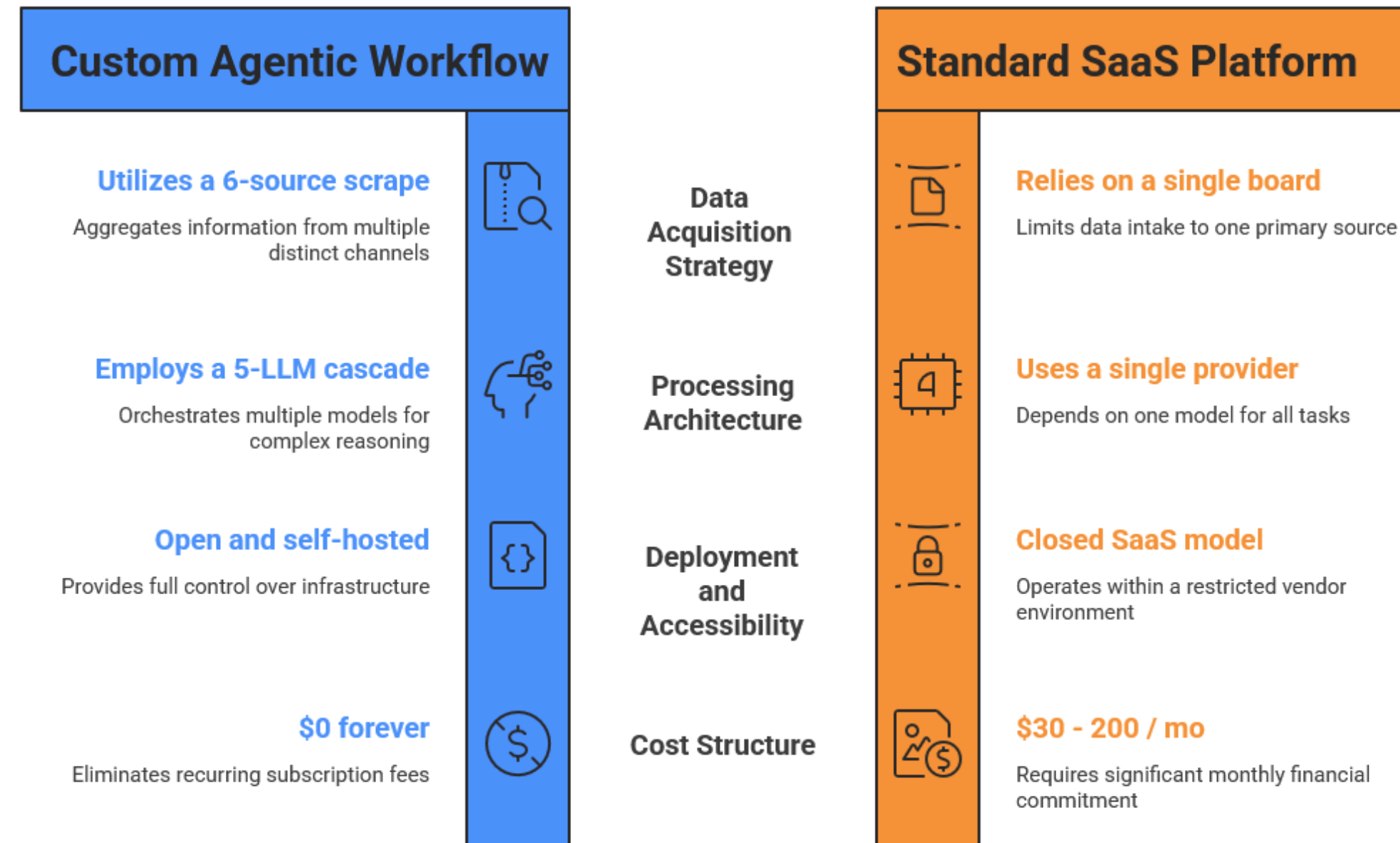
Why we stand out

- **Draft-only by default** - every competitor races to auto-send. We refuse to send without a green button.
- **Two surfaces, one state** - web and Telegram tap the same RLS-protected handler and broadcast the same Realtime event.
- **Cryptographic content binding** - payload_hash ties approval to the exact bytes the operator saw.
- **Open-source, self-hosted** - operator owns the agent, the data, the LLM keys.
- **\$0 budget production stack** - every layer sits on a sustainable free tier.

The 2 strong MOATs

- **Trust MOAT** - HITL + payload_hash makes us safe enough for real client outreach. No competitor can copy this without rewriting their core send loop.
- **Compounding-memory MOAT** - per-client memory.md grows every interaction. The longer an operator uses us, the smarter their pitches get - switching cost rises with usage.

Which automation strategy aligns with your operational requirements?



*"Drafts everything. Sends nothing.
Until you say yes."*

Selected Technical Achievements

- **Zero-budget production deployment** - every service used has a free tier sufficient for the prototype, including a 24/7 always-on agent runtime on Oracle Cloud Always Free.
- **Provider-agnostic LLM router** - 5 providers in cascade, automatic failover on rate limits, BudgetExceededError + USER_DAILY_BUDGET_USD guard, idempotency keys, response caching.
- **47 unit tests + 4 E2E specs** - Vitest across 7 files plus Playwright pitch-approve, pitch-reject, stale-hash 409 guard, signup-scout. CI fans tests across 3 packages in parallel.
- **Multi-source scraping** - parser-level fallbacks. JSON-LD JobPosting schema fallback specifically built for LinkedIn + Freelancer's anti-bot DOM scrambling.
- **Streaming pitch drafts** - server streams chunks via SSE, worker UPDATES pitches.draft + payload_hash every 5 chunks, dashboard renders chunks live via Realtime.
- **payload_hash content integrity** - SHA-256 binds approval to exact subject + body + lead_id. Mismatched hashes return 409 Conflict.
- **Two surfaces, one state** - web button and Telegram inline keyboard tap the same approval handler, hit the same RLS-protected table, broadcast the same Realtime event.
- **9 workspace skills + 11 MCP tools** - fully wired Gateway integration. Gateway runs on Oracle Cloud Always Free with auto-renewing Let's Encrypt HTTPS via Caddy + DuckDNS.
- **Stdio MCP bridge** - sidesteps Vercel serverless's incompatibility with long-lived SSE by tunnelling stdio MCP through the Gateway container to our HTTP-based /api/mcp endpoint.
- **Demo-mode parachute** - full read-only walkthrough state for a fresh user with zero LLM keys configured - judges always see something working.

By the numbers

- ~25K LOC - monorepo (3 apps + 5 packages)
- 17 SQL migrations - every change versioned
- 12 phases shipped over 3 days

Links & Submission

GitHub Repo

- github.com/Dheeran-git/openclaw-venture-partner

Live Demo (Web Dashboard)

Gateway (OpenClaw runtime)

- openclaw-vp.duckdns.org
- *Hosted on Oracle Cloud Always Free - Caddy + Let's Encrypt + DuckDNS.*

Documentation submitted

- PROJECT_OVERVIEW.md - this deck's source of truth (20 sections, fully grounded in shipped code).
- PRODUCTION_BUILD_GUIDE.md - authoritative build spec, definitions of done, ops rules.
- docs/HOSTING_GATEWAY.md - 3 zero-cost hosting paths.
- docs/RUNBOOK.md - incident response.

OpenClaw upstream

github.com/openclaw/openclaw
368k+ stars - daily release cadence - Peter Steinberger

Health endpoint

[/api/health](https://openclaw-vp.duckdns.org/api/health)
200/503 with database + LLM-provider checks.

Project window

2026-05-06 -> 2026-05-08
 12 phases - 10 complete - Phase 11 + 12
 Stage A done

Roadmap, Brand & Team

Roadmap - Near term (Phase 12 completion)

- Telegram + Discord webhooks fully migrated onto the Gateway (MCP-direct fallback active today).
- Slack channel activated (~30 min, Gateway-only).
- Standalone /api/telegram and /api/discord routes deleted (~700 LOC removal).

Medium term

- WhatsApp Business activation (after 3 - 10 days of Meta Business Verification).
- Daily scout cron migrated from Vercel cron to OpenClaw's cron tool.
- Browser tool re-enabled (requires VM upgrade 1 GB -> 4 GB).
- iOS / macOS companion apps bound to the Gateway URL (built into OpenClaw).

Long term

- Multi-tenant per-user Gateway provisioning - hosted-by-us model.
- Real customers, billing, paid Groq / Anthropic credits.
- ClawHub skill marketplace integration (publish our 9 skills).

Brand Identity

- **Mascot:** Molty the lobster (OpenClaw's official mark).
- **Primary color:** Coral #FF4D4D
- **Voice:** terse, builder-y, no emoji in product copy.
- **Tagline:** *"Drafts everything. Sends nothing. Until you say yes."*

Team - ClawGrowth

- **Lead:** Sathish
- **Affiliation:** RV College of Engineering, Bangalore
- **Built:** 2026-04-27 -> 2026-05-08

Thank you